Section A – Answer any 2 out of 3 questions in section A

Question 1

a) Why is the per-unit system used in analysis of power systems?

[4]

For the full mark, a short discussion around the following reasons should be listed:
Multiple voltage levels: 400 kV, 275 kV, 132 kV
Makes circuit analysis rather confusing
Impedance of transformers depends on side
Normalise all quantities to help understanding
Avoid confusion due to transformers.

b) A network, composed of a transformer and transmission line, supplies demand as shown in Figure 1a. Transformer rated primary and secondary voltages and apparent power are denoted as V_{R1}, V_{R2}, S_R, respectively. Line parameters are length, L, resistivity r' (Ω/km), reactivity x' (Ω/km), and susceptibility b' (S/km). Assuming apparent power base, S_{base}, and adequate voltage base value, state or derive the equations for each of the per-unit terms in Figure 1b (p_G, q_G etc.) based on the terms in Figure 1a.

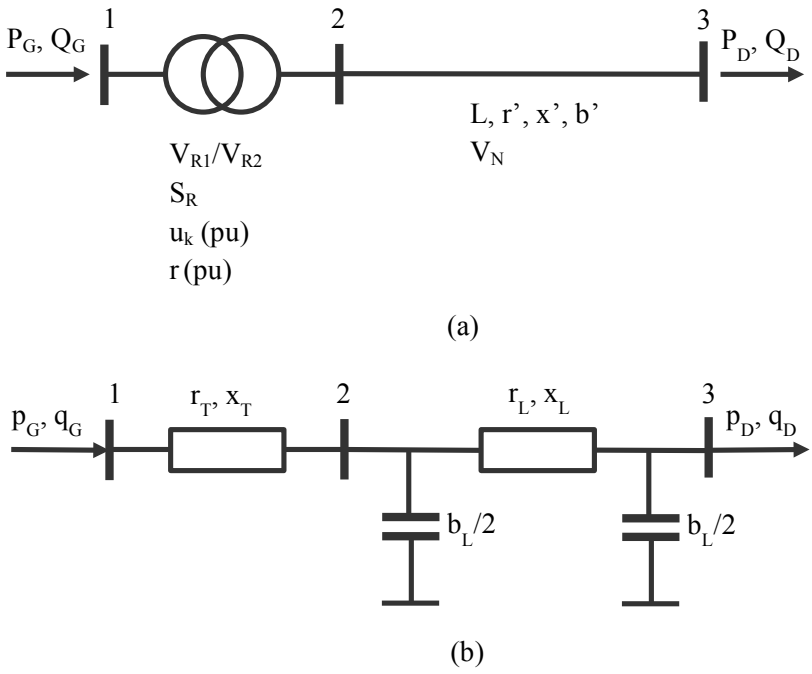


Figure 1: (a) Single line diagram of a network and (b) corresponding per unit representation

[8]

Generation and demand powers are

$$p_G = \frac{P_G}{S_{base}}$$

$$q_G = \frac{Q_G}{S_{base}}$$

$$p_D = \frac{P_D}{S_{base}}$$
$$q_D = \frac{Q_D}{S_{base}}$$

Voltage base is V_{R2} as transformer tap ratio is not be present in pu representation.

$$V_{base} = V_{R2}$$

Impedance base is

$$Z_{base} = \frac{V_{base}^2}{S_{base}}$$

Transformer parameters are

$$r_T = \frac{\left(r \cdot \frac{V_{R2}^2}{S_R}\right)}{Z_{base}} = r \cdot \frac{S_{base}}{S_R}$$
$$z_T = u_k \cdot \frac{S_{base}}{S_R}$$
$$x_T = \sqrt{z_T^2 - r_T^2}$$

Line parameters are

$$r_L = \frac{r'L}{Z_{base}} = r'L \cdot \frac{S_{base}}{V_{base}^2}$$
$$x_L = \frac{x'L}{Z_{base}} = x'L \cdot \frac{S_{base}}{V_{base}^2}$$
$$b_L = b'L \cdot Z_{base} = b'L \cdot \frac{V_{base}^2}{S_{base}}$$

c) Assuming S_{base}=100 MVA, for the transformer and overhead line shown in Figure 1 calculate the per unit parameters if the transformer parameters are 400/136 kV, 240 MVA, u_k=0.2 pu, r=0.05 pu and the 132 kV line parameters are L=50 km, r'=0.17 Ω/km, x'=0.38 Ω/km, b'=7.5 μS/km.

[4]

Transformer per unit parameters

$$S_{base} = 100\ MVA$$
$$V_{base} = 136\ kV$$
$$Z_{base} = \frac{136^2}{100} = 185\ \Omega$$
$$r_T = 0.05 \cdot \frac{100}{240} = 0.021\ pu$$
$$z_T = 0.2 \cdot \frac{100}{240} = 0.083\ pu$$
$$r_T = \sqrt{0.083^2 - 0.021^2} = 0.080\ pu$$

Line per unit parameters

$$r_L = 0.17 \cdot \frac{50}{185} = 0.046\ pu$$
$$x_L = 0.38 \cdot \frac{50}{185} = 0.103\ pu$$
$$b_L = 7.5 \cdot 10^{-6} \cdot 50 \cdot 185 = 0.069\ pu$$

d) A low voltage network supplies 100 households. Peak demand of each household is 10 kW and the coincidence factor for an infinite number of households is 0.1. In each household, a heat pump of 5 kW will be installed. If the coincidence factor of an infinite

number of heat pumps is 0.6 calculate the new coincidence factor for an infinite number of households assuming that peaks of households and heat pumps coincide. Estimate the new household coincidence factor for an infinite number of households. Perform a calculation assuming 100 households. Briefly explain the benefits of diversification.

[4]

The total diversified peak demand for n=100 households is

$$P_{100}^{a} = \mathrm{j}_{100}^{\mathrm{a}} 100\, P_1^a = \left(0.1 + \frac{1-0.1}{\sqrt{100}}\right) 100 \cdot 10 = 190\ kW$$

The total diversified peak demand for n=100 heat pumps is

$$P_{100}^{HP} = \mathrm{j}_{100}^{\mathrm{HP}} 100\, P_1^{HP} = \left(0.6 + \frac{1-0.6}{\sqrt{100}}\right) 100 \cdot 5 = 320\ kW$$

The total diversified peak demand for n=100 households when heat pumps are installed will be

$$P_{100}^{b} = P_{100}^{a} + P_{100}^{HP} = 190 + 320 = 510\ kW$$

Given that the new peak demand of a household will be 10+5=15kW, the new diversified peak can be expressed as

$$P_{100}^{b} = \mathrm{j}_{100}^{\mathrm{b}} 100\, P_1^b = \left(j_{\infty}^{b} + \frac{1-j_{\infty}^{b}}{\sqrt{100}}\right) 100 \cdot 15 = 510\ kW$$

Solving equation gives

$$j_{\infty}^{b} = 0.267$$

Diversification provides opportunity to supply demand with lower asset capacity then the total sum of demand. In this example, if there is no diversification the peak demand of 100 households with heat pumps will be 100 x 15 = 1500 kW i.e. three-fold compared to diversified peak.

Question 2

a) For a synchronous generator, derive equations for the active and reactive power delivered to the system if voltages, angle and impedance are known [3]

$$S = \sqrt{3}\bar{I_a}V_t = \sqrt{3}I_aV_t(cos\theta + jsin\theta)$$

$$\sqrt{3}I_acos\theta = \frac{E_isin\delta}{X_d}$$

$$P = \frac{|V_t||E_i|}{X_d}sin\delta$$

$$\sqrt{3}I_asin\theta = \frac{E_icos\delta - V_t}{X_d}$$

$$Q = \frac{|V_t|}{X_d}(|E_i|cos\delta - |V_t|)$$

b) Consider a three-phase synchronous generator, 200 MVA, 21 kV, connected to a very large system. Synchronous reactance of the generator is $X = 1.5$ p.u. with negligible armature resistance. Terminal voltage $V = 21$ kV, and the generator is supplying the active power of 150 MW at 0.90 power-factor (reactive power is supplied to the system). Find:

(i) Magnitude and phase angle of the current. [3]

The power factor angle can be obtained by $\theta = \cos^{-1} 0.9 = 25.84$ lagging
Current magnitude is

$$I_a = \frac{P}{\sqrt{3}V_tcos\theta} = \frac{150}{\sqrt{3}\cdot 21\cdot 0.9} = 4.582\ kA$$

(ii) Magnitude and phase angle of the synchronous internal voltage. [3]

Synchronous internal voltage is:

$$Z_{base} = \frac{21^2}{200} = 2.205\ \Omega$$
$$X_d = X\cdot Z_{base} = 1.5\cdot 2.205 = 3.3075\ \Omega$$

$$E_i = |E_i|(cos\delta + jsin\delta) = V_t + j\sqrt{3}X_dI_a = 21 + j\sqrt{3}\cdot 3.3075\cdot 4.582(0.9 - jsin25.84)$$
$$= 32.44 + j23.62 = 40.13(cos36.06 + jsin36.06)\ kV$$

(iii) Per unit magnitude of the synchronous internal voltage [2]

$$e_i = \frac{|E_i|}{V_R} = \frac{40.13}{21} = 1.91\ pu$$

(iv) Magnitude and angle between internal (*E*) and terminal voltage (*V*) if reactive power delivered to the system is doubled. [3]

Reactive power delivered to system is:

$$Q = P \cdot tan\theta = 150 \cdot \tan(25.84) = 72.64\ MVAr$$

Doubling reactive power

$$Q_2 = 2Q = 145.28\ MVAr$$

New angle between terminal voltage and current is

$$\theta_2 = \tan^{-1}\frac{145.28}{150} = 44.08$$

$$cos\theta_2 = 0.718$$

Current magnitude is

$$I_a = \frac{P}{\sqrt{3}V_t cos\theta} = \frac{150}{\sqrt{3} \cdot 21 \cdot 0.718} = 5.744\ kA$$

Synchronous internal voltage is:

$$E_i = |E_i|(cos\delta + jsin\delta) = V_t + j\sqrt{3}X_d I_a = 21 + j\sqrt{3} \cdot 3.3075 \cdot 5.744(0.718 - jsin44.08)$$
$$= 43.89 + j23.63 = 49.85(cos28.30 + jsin28.30)\ kV$$

c) Compute the probability that a system of 2 identical generators, each of 90 MW capacity and availability of 98%, and one wind farm of 30 MW and availability of 50%, will not meet peak demand of 100 MW.

[6]

Capacity (MW)	Probability	Cumulative probability
210	$98\%^2$ x 50% = 48.02%	100.00%
180	$98\%^2$ x (100% - 50%) = 48.02%	51.98%
120	2 x 98% x (100%-98%) x 50% = 1.96%	3.96%
90	2 x 98% x (100%-98%) x (100% - 50%) = 1.96%	2.00%
30	$(100\%-98\%)^2$ x 50% = 0.02%	0.04%
0	$(100\%-98\%)^2$ x (100% - 50%) = 0.02%	0.02%

Probability that demand will not be met is 2.06%.

Question 3

A three-bus power network is presented in Figure 3 below. Data relevant for the load flow analysis on this system are given in per unit values.

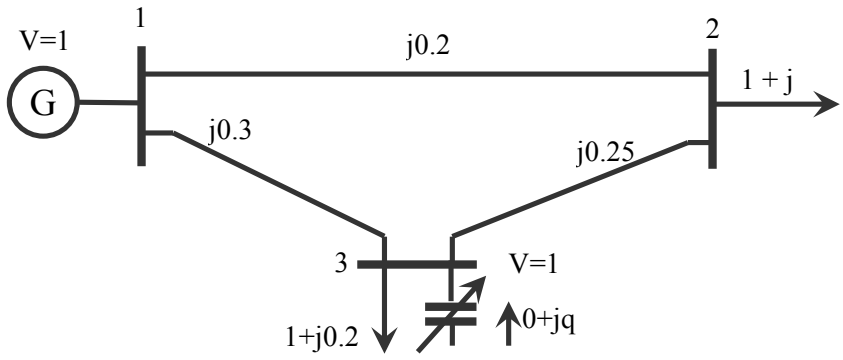


Figure 3. Three bus network showing per unit data for voltages,

active and reactive loads and line reactances

a) Explain the concept of the PV bus in the load flow calculations

[2]

It denotes bus where active power injection and voltage (magnitude) remain constant in power flow calculation. Reactive power is controlled to keep desired voltage (magnitude).

b) Form the Ybus matrix and explain why it is singular (i.e. not invertible)

[2]

The Ybus is

$$Y = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} -j5 & 0 & 0 \\ 0 & -j3.333 & 0 \\ 0 & 0 & -j4 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} -j8.333 & j5 & j3.333 \\ j5 & -j9 & j4 \\ j3.333 & j4 & -j7.333 \end{bmatrix}$$

Matrix is singular as there are N-1 independent equations I = Y V.

c) Perform 2 iterations of the Gauss-Seidel load flow method

[9]

The first iteration of Gauss Siedel is as follows.

It is $V_1^{(0)} = V_2^{(0)} = V_3^{(0)} = 1 + j0$ where 1: slack bus, 2: PQ bus, 3: PV bus.

$$V_2^{(1)} = \frac{1}{Y_{22}} \cdot \left(\frac{S_2^{(0)*}}{V_2^{(0)*}} - Y_{21} \cdot V_1^{(0)} - Y_{23} \cdot V_3^{(0)} \right) = \frac{1}{-j9} \cdot \left(\frac{-1 + j1}{1} - j5 \cdot 1 \; - j4 \; \cdot 1 \right)$$

$$= (0.8889 - j0.1111) \text{ p. u.}$$

Tentatively calculate voltage at PV bus 3:

$$\tilde{V}_3^{(1)} = \frac{1}{Y_{33}} \cdot \left(\frac{S_3^{(0)*}}{V_3^{(0)*}} - Y_{31} \cdot V_1^{(0)} - Y_{32} \cdot V_2^{(1)} \right)$$

$$= \frac{1}{-j7.333} \cdot \left(\frac{-1}{1} - j3.333 \cdot 1 \ - j4 \cdot (0.8889 - j0.1111) \right)$$

$$= (0.9394 - j0.197)\ p.u.$$

Update voltage at PV bus 3 to specified magnitude while keeping angle constant:

$$V_3^{(1)} = V_3^{spec} \cdot \frac{\tilde{V}_3^{(1)}}{\left|\tilde{V}_3^{(1)}\right|} = 1.00 \ \cdot \frac{0.9394 - j0.197}{|0.9394 - j0.197|} = (0.9787 - j0.2052)\ \text{p. u.}$$

At the end of an iteration we calculate reactive power injected at PV bus 3, Q_3, in order to update it with the new values for voltage. Alternative approach is to update Q before updating voltage at PV bus.

$$Q_3^{(1)} = -Im\left(V_3^{(1)*} \cdot \left(Y_{31} \cdot V_1^{(0)} + Y_{32} \cdot V_2^{(1)} + Y_{33} \cdot V_3^{(1)} \right) \right)$$

$$= -Im\Big((0.9787 + j0.2052) \cdot \big(j3.333 \ \cdot 1 + j4 \ \cdot (0.8889 - j0.1111) + (-j7.333) \cdot (0.9787 - j0.2052) \big) \Big) = 0.4999\ p.u.$$

$$S_3^{(1)} = \ Re\left(S_3^{(0)} \right) + jQ_3^{(1)} = (-1 + j0.4999).\, p.u.$$

Second iteration

$$V_2^{(2)} = \frac{1}{Y_{22}} \cdot \left(\frac{S_2^{(1)*}}{V_2^{(1)*}} - Y_{21} \cdot V_1^{(0)} - Y_{23} \cdot V_3^{(1)} \right)$$

$$= \frac{1}{-j9} \cdot \left(\frac{-1 + j1}{0.8889 + j0.1111} - j5 \cdot 1 \ - j4 \ \cdot (0.9787 - j0.2052) \right)$$

$$= (0.8521 - j0.1989)\ \text{p. u.}$$

$$\tilde{V}_3^{(2)} = \frac{1}{Y_{33}} \cdot \left(\frac{S_3^{(1)*}}{V_3^{(1)*}} - Y_{31} \cdot V_1^{(0)} - Y_{32} \cdot V_2^{(2)} \right)$$

$$= \frac{1}{-j7.333} \cdot \left(\frac{-1 - j0.4999}{0.9787 + j0.2052} - j3.333 \cdot 1 \ - j4 \cdot (0.8521 - j0.1989) \right)$$

$$= (0.958 - j0.2559)\ \text{p. u.}$$

$$V_3^{(2)} = V_3^{spec} \cdot \frac{\tilde{V}_3^{(2)}}{\left|\tilde{V}_3^{(2)}\right|} = 1.00 \ \cdot \frac{0.958 - j0.2559}{|0.958 - j0.2559|} = (0.9661 - j0.2581)\ \text{p. u.}$$

$$Q_3^{(2)} = -Im\left(V_3^{(2)*} \cdot \left(Y_{31} \cdot V_1^{(0)} + Y_{32} \cdot V_2^{(2)} + Y_{33} \cdot V_3^{(2)} \right) \right)$$

$$= -Im\Big((0.9661 + j0.2581) \cdot \big(j3.333 \ \cdot 1 + j4 \ \cdot (0.8521 - j0.1989) + (-j7.333) \cdot (0.9661 - j0.2581) \big) \Big) = 0.6148\ \text{p. u.}$$

$$S_3^{(2)} = \ Re\left(S_3^{(0)} \right) + jQ_3^{(2)} = (-1 + j0.6148)\ p.u.$$

d) Using the results from c), calculate the power mismatch at the PQ bus

[4]

The mismatch is found as follows:
Flows from PQ bus

$$S_{21} = V_2^{(0)} \cdot \left(\left(V_2^{(0)} - V_1^{(0)}\right) \cdot y_{21}\right)^*$$
$$= (0.8521 - j0.1989) \cdot \left(\left((0.8521 - j0.1989) - 1\right) \cdot (-j5)\right)^*$$
$$= (-0.9945 - j0.4324) \text{ p. u.}$$
$$S_{23} = V_2^{(2)} \cdot \left(\left(V_2^{(2)} - V_3^{(2)}\right) \cdot y_{23}\right)^*$$
$$= (0.8521 - j0.1989)$$
$$\cdot \left(\left((0.8521 - j0.1989) - (0.9661 - j0.2581)\right) \cdot (-j4)\right)^*$$
$$= (0.111 - j0.4358) \text{ p. u.}$$

Power mismatch in PQ bus
$S_2 = S_{21} + S_{23} = -0.8835 - j0.8682\ p.u.$
$M_2 = S_2^{spec} - S_2 = (-1 - j) - (S2\ ) = -0.1165 - j0.1318\ p.u.$

e) Calculate the reactive power from compensator at the PV bus

[3]

$$q^{(2)} = Q_3^{(2)} - (-0.2) = 0.6148 + 0.2 = 0.8148 \text{ p. u.}$$

Section B – Answer any 2 out of 3 questions in section B

Question 4

a) Short circuit calculations may be simplified by neglecting the pre-fault currents and assuming the pre-fault voltages to be 1.0 p.u. Justify use of these assumptions in the context of high voltage bulk power transmission system.

[4]

Assumption 1: Pre-fault voltages are 1.0 pu
In practice under normal operation voltages are within ±5% of 1.0 pu. So assuming the pre-fault voltages to be 1.0 p.u. is a fair assumption

Assumption 2: Pre-fault currents are zero
- change in current due to fault is quite large, typically 10-20 times
- fault current is approximately in phase quadrature with respect to the pre-fault current
- hence the total current (pre-fault + during fault) magnitude can be assumed to be the larger of the two justifying zero pre-fault current assumption

[1 mark each for pre-fault voltage and the arguments justifying zero pre-fault current]

b) What are the four steps for performing balanced fault current calculation without any simplifying assumptions about pre-fault voltages and currents.

[4]

1. Obtain the pre-fault steady state solution V_0 and I_0 from power flow analysis

2. Replace the reactance of synchronous machines (generator and motor) by their sub-transient values

3. Excite the passive Thevenin equivalent (emf sources shorted) at the fault point by the negative of the pre-fault voltage in series with fault impedance and compute the sub-transient voltage and current at all points of interest

4. Fault current is obtained by adding the results of steps 1 (pre-fault) and 3 (sub-transient): Superposition

[1 mark for each step]

c) Explain, using the theorem of constant flux linkage, why there is a decaying AC component in the stator current immediately after a three-phase short circuit on a generator terminal.

[5]

- Due to a three-phase fault on the generator terminal, the stator current and hence, the magnetic flux increases but according to the constant flux linkage theorem, flux linking a closed winding cannot change instantaneously.
- Therefore, the magnetic flux due to short-circuit stator current is forced through high reluctance paths that do not link the field (rotor) or damper circuits.
- Thus, the stator inductance which is inversely proportional to reluctance is initially low leading to high current.

- As the flux penetrates the damper and field circuit which are lower reluctance paths, stator inductance increases resulting in reduced currents in steady-state compared to the transient or sub-transient conditions.

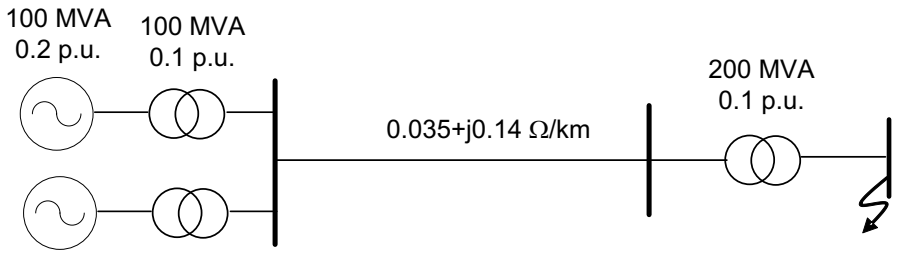
  This explains the decaying AC component in the stator current.

  [1 mark for each point and 1 mark for coherence in argument]

d) Two 100 MVA, 20 kV three-phase generators are connected to a 132 kV busbar through their individual generator transformer. The sub-transient reactance of both generators is 0.2 p.u. and the leakage reactance of both transformers is 0.1 p.u. From the 132 kV busbar, a 40 km long 132 kV line supplies an 11 kV load through a 200 MVA, 132/11 kV transformer with a leakage reactance of 0.1 p.u. The line impedance per phase is (0.035 +*j*0.14) Ω/km. Using a 100 MVA base, calculate the following for a balanced three-phase short circuit on the low-voltage terminals of the load transformer:

(i) fault current (in kA) through the 132 kV line.

[5]



On 100 MVA base

$$Z_{base} = \frac{132^2}{100} = 174.24\ \Omega$$

$$I_{base} = \frac{100}{\sqrt{3}\times 132} = 0.437\ \text{kA}$$

Line impedance in p.u. is $\frac{0.035+j0.14}{174.24} = 0.008 + j0.032$ p.u.

Equivalent impedance at the fault point is

$$Z_{eq} = j\frac{0.2+0.1}{2} + (0.008 + j0.032) + j0.1\times\frac{100}{200} = 0.008 + j0.232\ \text{p.u.}$$

$$|Z_{eq}| = 0.2321\ \text{p.u.}$$

Fault current is

$$I_f = \frac{1}{0.2321} = 4.308\ \text{p.u.} = 4.308\times I_{base} = 1.883\ \text{kA}$$

[1 mark for base calculation, 3 marks for impedance calculation, 1 mark for fault current]

(ii) rating of the circuit breaker required at the load end of the 132 kV line. [2]

Rating of the circuit breaker is

$$S_{\text{CB}} = \frac{MVA_{\text{Base}}}{|Z_{\text{eq}}|} = \frac{100}{0.2321} = 430.85 \text{ MVA}$$

[1 mark each for correct expression and calculation/units]

Question 5

a) Starting from the fault current and voltage in phase domain explain how the three sequence networks should be connected for analysing the following unbalanced faults:

(i) line to line to ground (LLG) fault affecting phases B and C with a fault impedance Z_f to ground

[5]

Fault current and voltage in phase domain:
$I_A = 0, \quad V_B = V_C = I_f Z_f = (I_B + I_C) Z_f$

Fault current and voltage in sequence domain:
$I_A = I_{A0} + I_{A1} + I_{A2} = 0$

$I_f = I_B + I_C = 2I_{A0} + (\alpha + \alpha^2)(I_{A1} + I_{A2}) = 3I_{A0}$

$$\begin{bmatrix} V_{A0} \\ V_{A1} \\ V_{A2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}\begin{bmatrix} V_A \\ V_B \\ V_B \end{bmatrix}$$

$V_{A1} = V_{A2}$ and $V_{A0} - V_{A1} = \frac{1}{2}(2 - \alpha - \alpha^2)V_B = V_B = 3Z_f I_{A0}$

Thus, for LLG fault, the positive and negative sequence networks are connected in parallel with 3 times the fault impedance between positive/negative and zero sequence networks.

[1 mark for phase domain expressions, 3 marks for deriving voltage and current in sequence domain, 1 mark for inferring the right circuit connection]

(ii) line to ground (LG) fault on phase A with a fault impedance Z_f to ground

[5]

Fault current and voltage in phase domain:
$I_B = I_C = 0, \quad V_A = I_A Z_f$

Fault current and voltage in sequence domain:

$$\begin{bmatrix} I_{A0} \\ I_{A1} \\ I_{A2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}\begin{bmatrix} I_A \\ 0 \\ 0 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} I_A \\ I_A \\ I_A \end{bmatrix}$$

$I_{A0} = I_{A1} = I_{A2} = \frac{1}{3} I_A$

$V_A = V_{A0} + V_{A1} + V_{A2} = I_A Z_f = 3I_{A0} Z_f$

Thus, for LG fault, the positive, negative and zero sequence networks are connected in series with 3 times the fault impedance.

[1 mark for phase domain expressions, 3 marks for deriving voltage and current in sequence domain, 1 mark for inferring the right circuit connection]

b) A 66 kV, three-phase transmission line is supplied by a 11 kV generator through a 11/66 kV transformer having star connection with grounded neutral on both primary and secondary sides. The positive-sequence reactance of the transformer and the transmission line are $j10.9\ \Omega$ and $j44\ \Omega$, respectively. The equivalent positive and negative-sequence sub-transient reactance of the generator, referred to the line voltage of 66 kV, are $j18\ \Omega$ and $j14.5\ \Omega$, respectively. The equivalent zero sequence reactance measured from the generator up to the fault point is $j150\ \Omega$ referred to the line voltage of 66 kV. For faults at the load end of the 66 kV transmission line, calculate the following neglecting the fault impedance and any resistance. Use 100 MVA base for the calculations.

(i) Fault current (in kA) due to a line to ground (LG) fault on phase A.

[5]

$S_{\text{Base}} = 100\ \text{MVA}$

On 66 kV side,

$$Z_{\text{Base}} = \frac{66^2}{100} = 43.56\ \Omega$$

$$I_{\text{Base}} = \frac{100}{\sqrt{3}\times 66^2} = 0.875\ \text{kA}$$

Per unit value of the positive, negative and zero sequence reactance of the generator, transformer and line are

Component	Positive sequence reactance (p.u.)	Negative sequence reactance (p.u.)	Zero sequence reactance (p.u.)
Generator	$\frac{18}{43.56} = 0.413$	$\frac{14.5}{43.56} = 0.333$	$\frac{150}{43.56} = 3.444$
Transformer	$\frac{10.9}{43.56} = 0.25$		
Transmission Line	$\frac{44}{43.56} = 1.01$		
Total reactance	$0.413+0.25+1.01$ $=1.673$ p.u.	$0.333+0.25+1.01$ $= 1.593$ p.u.	3.444 p.u.

LG fault current is

$$I_{\text{f-LG}} = \frac{3}{1.673 + 1.593 + 3.444} = 0.447\ \text{p.u.} = 0.447 \times I_{\text{Base}} = 0.391\ \text{kA}$$

[1 mark each for base calculation, determining positive, negative and zero sequence impedances in per unit and fault current calculation]

(ii) Current in phase B (in kA) during a line to line to ground (LLG) fault affecting phases B and C.

[5]

For a LLG fault, the positive, negative and zero sequence currents are:

$$I_{A1} = \frac{1}{1.673 + \frac{1.593 \times 3.444}{1.593 + 3.444}} = 0.362 \text{ p.u.}$$

$$I_{A2} = -0.362 \times \frac{3.444}{1.593 + 3.444} = -0.248 \; p.u.$$

$$I_{A0} = -0.362 \times \frac{1.593}{1.593 + 3.444} = -0.114 \; p.u.$$

Current in phase B is

$I_{f-B} = -0.114 + \alpha^2 \times 0.362 + \alpha \times (-0.248) = -0.171 - j0.528$ p.u.

$\left|I_{f-B}\right| = 0.555$ p.u. $= 0.555 \times I_{\text{Base}} = 0.486$ kA

[1 mark each calculation of positive, negative and zero sequence currents and 2 marks for Phase B current in kA]

Question 6

a) Consider the standard power angle characteristic for a round rotor generator with zero resistance. Explain physically why the segment of the power angle characteristic with a positive slope corresponds to stable operation in steady state while that with a negative slope represents instability.

[4]

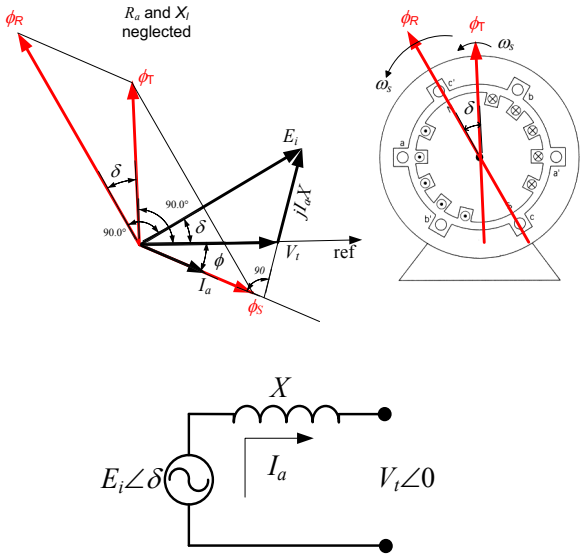
- If the operating point is in the positive slope region, any increase in mechanical input leads to acceleration of the rotor resulting in increase in load angle and hence electrical power output. Thus increase/decrease in mechanical power input implies corresponding increase/decrease of electrical power output which helps maintain stability.

- On the other hand, if the operating point is in the negative slope region, any increase in mechanical input results in increase in load angle as before, but on this occasion increase in load angle leads to reduction in electrical power output because of the negative slope. Thus, an increase in mechanical input results in reduction of electrical power output and making the situation is unstable.

[2 marks each for stable and unstable operating points]

b) "The active power delivered by a round rotor synchronous generator depends on the angular separation between the rotor and the total magnetic field produced by the stator and the rotor windings". Justify this statement using a phasor diagram. Neglect any resistance of the generator.

[5]



From the phasor diagram, it is clear that the angular separation between the field (ϕ_R) and the resultant magnetic field (ϕ_T) given by δ is the same as the angular difference between the internal emf (E_i) and terminal voltage (V_t).

Active power delivered by the synchronous generator is given by:

$$P = \frac{E_i V_t}{X} \sin\delta$$

Hence, the active power delivered by a round rotor synchronous generator is decided by the angular separation between the rotor and the total magnetic field produced by the stator and the rotor windings.

[2 marks for correct phasor diagram, 2 marks for relating the magnetic fields to the voltages, 1 mark for coherence in argument]

c) A round-rotor synchronous generator with a synchronous reactance of 1.0 p.u. is connected to two parallel lines through a step-up transformer with 0.1 p.u. leakage reactance. The reactance of each line is 0.4 p.u. At the other end of the lines a step-down transformer with 0.1 p.u. leakage reactance supplies a synchronous motor on the LV side. The motor is of the round-rotor type and has a synchronous reactance of 1.0 p.u. The internal voltage of the generator is 1.2 p.u. and of the motor 1.0 p.u. If the generator produces 0.3 p.u. active power, determine the critical clearing angle for the following fault condition on one of the lines. Neglect the resistances and assume the power transfer between the generator and the motor to be zero during the fault.

(i) when the line fault is temporary and is cleared by itself.

[6]

In pre-fault condition, the total reactance between the generator and the motor is

$$X_T = 1.0 + 0.1 + \frac{0.4}{2} + 0.1 + 1 = 2.4 \text{ p.u.}$$

Steady-state stability limits is:

$$P_{max} = \frac{E_G \times E_M}{\mathrm{X_T}} = \frac{1.2 \times 1}{2.4} = 0.5 \text{ p.u.}$$

Power angle is given by

$$\sin\delta_0 = \frac{P}{P_{max}} = \frac{0.3}{0.5} \rightarrow \delta_0 = 36.87°$$

Maximum allowable power angle is

$$\delta_{\max} = 180° - \delta_0 = 180° - 36.87° = 143.13°$$

Accelerating area is

$$\int_{\delta_0}^{\delta_{\mathrm{cl}}} (P - 0)d\delta = 0.3 \times (\delta_{\mathrm{cl}} - 36.87°) \text{ p.u.-deg}$$

Decelerating area is

$$\int_{\delta_{\mathrm{cl}}}^{\delta_{\max}} (P_{\max} \sin\delta - P)d\delta$$

$$=$$

$$\int_{\delta_{cl}}^{\delta_{max}} 0.5\times(\cos\delta_{cl} - \cos 143.13°) + 0.3\times(\delta_{cl} - 143.13°) \text{ p.u.-deg}$$

Using equal-area criteria

$$\delta_{cl} = \cos^{-1}\left(\frac{0.5\times\cos 143.13° + 0.3\times143.13\times{}^{\pi}/_{180} - 0.3\times36.87\times{}^{\pi}/_{180}}{0.5}\right)$$
$$= 71.77°$$

[1 mark each for steady-state stability limit, initial power angle, maximum power angle, accelerating and decelerating area and critical clearing angle]

(ii) when the line fault is permanent and is cleared by taking the faulty line out of service.

[5]

In post-fault condition the total reactance between the generator and the motor is

$X_T = 1.0 + 0.1 + 0.4 + 0.1 + 1 = 2.6$ p.u.

Post-fault steady-state stability limits is:

$$P_{max2} = \frac{E_G\times E_M}{X_T} = \frac{1.2\times1}{2.6} = 0.46 \text{ p.u.}$$

Power angle is given by

$$\sin\delta_f = \frac{P}{P_{max2}} = \frac{0.3}{0.46} \rightarrow \delta_f = 40.54°$$

Maximum allowable power angle is

$$\delta_{max} = 180° - \delta_f = 180° - 40.54° = 139.46°$$

Accelerating area is

$$\int_{\delta_0}^{\delta_{cl}} (P - 0)d\delta = 0.3\times(\delta_{cl} - 36.87°) \text{ p.u.-deg}$$

Decelerating area is

$$\int_{\delta_{cl}}^{\delta_{max}} (P_{max2}\sin\delta - P)d\delta$$

$$= \int_{\delta_{cl}}^{\delta_{max}} 0.46\times(\cos\delta_{cl} - \cos 139.46°) + 0.3\times(\delta_{cl} - 139.46°) \text{ p.u.-deg}$$

Using equal-area criteria

$$\delta_{cl} = \cos^{-1}\left(\frac{0.46\times\cos 139.46° + 0.3\times 139.46\times {}^{\pi}/_{180} - 0.3\times 36.87\times {}^{\pi}/_{180}}{0.46}\right) = 66.02°$$

[1 mark each for post-fault power angle, maximum power angle, accelerating and decelerating area and critical clearing angle]